

The Field-Tensor Dictionary: From the Mechanical State of a Tensioned Two-Strand Rope to Electromagnetism

July 2026. The rigorous assembly of the corpus's electromagnetic identifications into a single mapping: mechanical state $\rightarrow$ potentials $\rightarrow$ field tensor $\rightarrow$ forces, with every line either Derived or explicitly flagged, and both force-sign rules obtained from mechanics with nothing put in by hand.

1. The mechanical state variables

The two-strand rope under tension T_0 carries one internal gapless degree of freedom: the braid phase Φ , the single Goldstone mode of the internal symmetry (the screw generator; the one-generator theorem, GRV-020). Its derivatives are the physically meaningful densities: the TWIST DENSITY $\eta = \partial\Phi/\partial s$ (structural handedness per unit length) and the ROTATION RATE $\omega = \partial\Phi/\partial t$ (the helix turning like a screw). The tension channel couples to twist through the twist-stretch lock, $k_{tw} = k + \lambda\gamma^2\tau_0^2$ with $\gamma = 1/\sin^2\theta$ derived from helix geometry (EM-RECON-012). Coarse-graining the strand orientations over many ropes yields the orientation field, carrying a connection in the fiber-bundle sense (GG-001).

2. The dictionary

Electromagnetic object	Mechanical identity	Status	Claims
Charge q	Winding / linking number of the two strands: an integer; handedness is the sign	Derived (quantized)	GG-006, EM-001
Current J	Transported linking: the helix turned like a screw; one turn = one linking unit past each cross-section; $\text{div } J = 0$ is the no-wind-up condition	Derived	EM-008, EM-014
Vector potential A	The coarse-grained orientation connection; gauge freedom = phase-convention arbitrariness (physically: rigid braid-phase relabeling, the GRV-020 symmetry)	Derived (framework)	GG-001, GG-004
Scalar potential φ	The twist-tension channel: stored twist transmitted to tension by the twist-	Modeled	EM-RECON-011, -012

	stretch lock		
$E_i = F_{i0} = -\partial_i \phi - \partial_t A_i$	Static part: along-rope tension/twist gradient. Induction part: time-varying orientation transport (Faraday, inside $dF = 0$)	Assembled here	EM-RECON-011, EM-003
$B_i = \frac{1}{2} \epsilon_{ijk} F_{jk}$	Curl of the orientation field: the surrounding network's circulating response to the rotation rate ω	Derived / Modeled	EM-009, EM-010, EM-014
Field magnitude	$g = E\sqrt{(\epsilon_0/\Sigma)}$: strain per field, with ONE constant $\Sigma = T_{on_L}$ (vacuum tension density), bounded by ATLAS/PVLAS	Modeled (calibrated form)	EM-RECON-014, -016
Maxwell equations	$dF = 0$ (Bianchi identity) + $d * F = J$ (Chern–Weil, $d = 3$)	Derived	EM-003
Lorentz force $q(E + v \times B)$	From the coupling $q v \cdot a$, FORCED by gauge invariance + current conservation; correct coefficient; $\perp$ to v and B ; does no work	Derived	EM-013, EM-010
Magnetic force sign	Gap geometry: same-sense circulations produce opposite swing velocities between wires $\rightarrow$ mesh $\rightarrow$ attract; opposite $\rightarrow$ clash $\rightarrow$ repel	Derived	EM-012
Electrostatic force sign	Constraint-source superposition: like twist fields ADD between knots (repel); opposite CANCEL (attract); $+q_1 q_2 / (4\pi K d)$	Derived (new)	EM-015

3. The electrostatic sign theorem (the new result)

The trap, named first: a quadratic field between sources does NOT automatically repel like charges. With COUPLING sources (an energy term $-\rho\Phi$, the structure of scalar exchange), the minimized interaction is $-q_1 q_2 / (4\pi K d)$: like charges attract — which is why scalar-mediated forces attract and why naive arguments get Coulomb backwards. The rope winding is not a

coupling source. Winding number is TOPOLOGY: a boundary condition on the braid-phase configuration space, with nothing to couple and nothing to switch off. For such CONSTRAINT sources the energy is purely $(K/2)|\nabla\Phi|^2$ on the constrained minimizer, and superposition gives $U = +q_1q_2/(4\pi Kd)$: like windings repel, opposite attract, with the $1/d$ potential and $1/d^2$ force of Coulomb's law. The laboratory precedent is exact: superfluid vortices are constraint-sourced quadratic fields, and like-sign vortices repel while vortex–antivortex pairs attract. Electron and positron are opposite handedness (GG-006): opposite winding, hence attraction. The benchmark verifies both the analytic dichotomy (the two source types provably carry opposite signs, so the trap is permanently fenced) and the mechanism on a lattice of actual winding defects: the like pair's energy falls with separation (repulsion), the opposite pair's rises (attraction). In plain terms: like-handed twist fields add in the region between the knots, storing more energy when they are close, so they are pushed apart; opposite-handed fields cancel between the knots, so they are pulled together. With the magnetic half (the gap-geometry rule of EM-012), the sign ledger of electromagnetism is now closed entirely by geometry.

4. The status ledger: why Modeled and not Derived

The dictionary is assembled with its remaining debts explicit. (i) Σ 's absolute value is an input: measured and bounded by experiment (ATLAS; PVLAS), identified in form as T_{on_L} , not derived from within. (ii) The dynamical drive's inertial term (EM-010) is assumed, not derived. (iii) The scalar potential's channel identification (EM-RECON-011) is Modeled. (iv) The identification's UNIQUENESS — that no other assignment of mechanical variables reproduces the same phenomenology — is not established. On this assembly the corpus's central identification claim, GG-005 ('F IS the electromagnetic field; topology IS charge'), is upgraded from Conjecture to Modeled: supported by structure, quantization, forces with both signs, calibration, and dynamics; withheld from Derived by exactly the four debts above.

5. What would strengthen each line

Deriving Σ from the mesh scale and tension (blocked at present by the same irreducibility results that block a and $\hbar$); deriving EM-010's inertial term from strand dynamics; promoting the φ -channel identification by an independent prediction in the static sector; and a uniqueness argument — or a counterexample assignment, which would be equally valuable and would go into the registry the same way everything else has.